Basics of Parametric Equations

Exam code: 9MA0

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Written by: Paul
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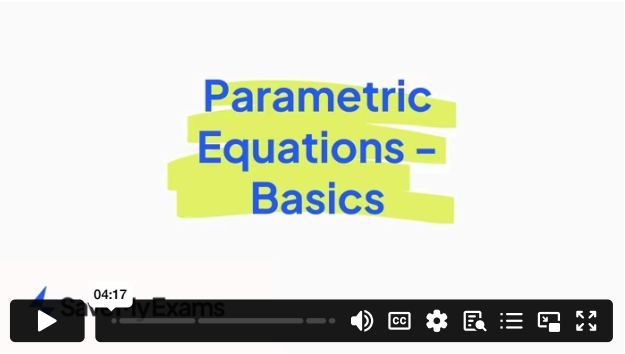

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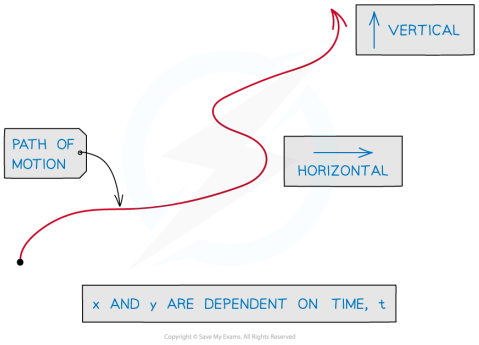
Help me understand

Basics of parametric equations

What are parametric equations?

- Graphs are usually described by a **Cartesian equation**
  - The equation involves **x** and **y** only
- Equations like this can sometimes be rearranged into the form, **y = f(x)**



- In **parametric equations** both **x** and **y** are dependent on a third variable
  - This is called a **parameter**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **y** is the vertical position of an object
- and the position of the object is dependent on time **t**

- **x** is a function of **t**, **y** is a function of **t**
  - **x = f(t)**
  - **y = g(t)**

How do I plot parametric equations?

- It is still possible to plot a graph of **y** against **x** from their parametric equations

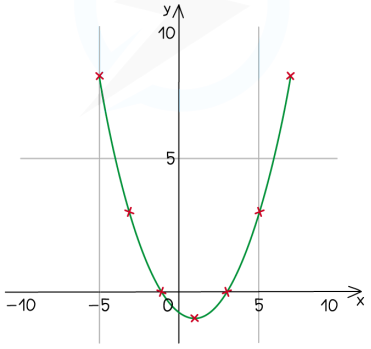
PLOTTING A GRAPH FROM PARAMETRIC EQUATIONS

e.g. PLOT THE GRAPH GIVEN BY THE PARAMETRIC EQUATIONS $x = 2t + 1$ AND $y = t^2 - 1$ FOR $-3 \leq t \leq 3$.

CONSTRUCT A TABLE OF VALUES

t	−3	−2	−1	0	1	2	3
x	−5	−3	−1	1	3	5	7
y	8	3	0	−1	0	3	8

PLOT THE GRAPH OF y AGAINST x



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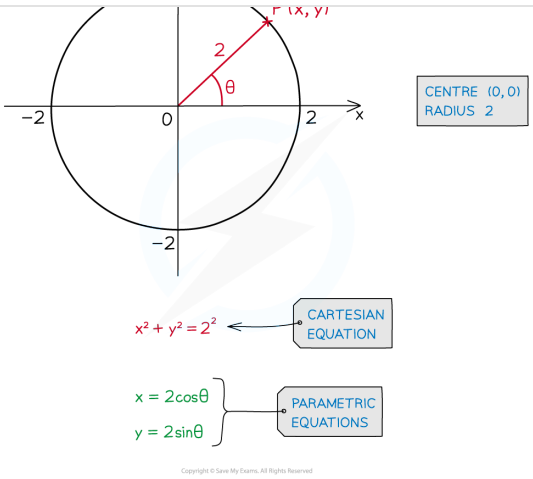
- Also see Parametric Equations – Sketching Graphs

What are the parametric equations of a circle?



- For a circle, centre (0, 0) and radius **r**
  - **x = rcos θ**
  - **y = rsin θ**
  - (Note that **r** is constant, this is not two parameters)
- For a circle, centre (**a**, **b**) and radius **r**
  - **x = rcos θ + a**
  - **y = rsin θ + b**

Worked Example

(a) Write down the radius and the centre of the circle defined by the parametric equations

$$x = 3\cos\theta$$

and

$$y = 3\sin\theta - 4$$

(b) Hence write down the Cartesian equation of the circle

a) CENTRE: (0, −4)
RADIUS: 3

$x = r\cos\theta + a$
$y = r\sin\theta + b$

b) $x^2 + (y + 4)^2 = 9$

$r^2 = 3^2 = 9$

CENTRE (0, −4)

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